



Graphs in Machine Learning

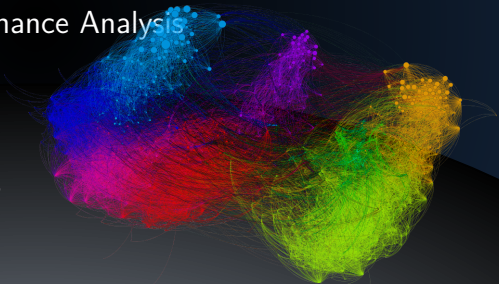
Analysis of Online SSL

Quantization Error and Performance Analysis

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Partially based on material by: Branislav Kveton,
Mikhail Belkin, Jerry Zhu



Online SSL with Graphs: Analysis

Want to bound $\frac{1}{N} \sum_{t=1}^N (\hat{f}_{soq,t}[t] - y_t)^2$

What can we guarantee?

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- generalization error — if all data: $(\hat{f}_{s,t} - y_t)^2$
- online error — data only incrementally: $(\hat{f}_{so,t}[t] - \hat{f}_{s,t})^2$
- quantization error — memory limitation: $(\hat{f}_{soq,t}[t] - \hat{f}_{so,t}[t])^2$

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All together:

$$\frac{1}{N} \sum_{t=1}^N (\hat{f}_{soq,t}[t] - y_t)^2 \leq \frac{9}{2N} \sum_{t=1}^N (\hat{f}_{s,t} - y_t)^2 + \frac{9}{2N} \sum_{t=1}^N (\hat{f}_{so,t}[t] - \hat{f}_{s,t})^2 + \frac{9}{2N} \sum_{t=1}^N (\hat{f}_{soq,t}[t] - \hat{f}_{so,t}[t])^2$$

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Bounding **transduction error** $(\hat{f}_{s,t} - y_t)^2$

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If all labeled examples l are i.i.d., $c_l = 1$ and $c_l \gg c_u$, then

$$R(\hat{\ell}^*) \leq \hat{R}(\hat{\ell}^*) + \underbrace{\beta + \sqrt{\frac{2 \ln(2/\delta)}{n_l}} (n_l \beta + 4)}_{\text{transductive error } \Delta_T(\beta, n_l, \delta)}$$
$$\beta \leq 2 \left[\frac{\sqrt{2}}{\gamma_g + 1} + \sqrt{2n_l} \frac{1 - c_u}{c_u} \frac{\lambda_M(\mathbf{L}) + \gamma_g}{\gamma_g^2 + 1} \right]$$

holds with the probability of $1 - \delta$, where

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$\beta = O(n_l^{-(1+\alpha)})$ for any $\alpha > 0$

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Bounding **online error** $(\hat{f}_{so,t}[t] - \hat{f}_{s,t})^2$

Idea: If $\mathbf{L}$ and $\hat{L}_o$ are regularized, then HFSs get closer together.

since they get closer to zero

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and also $\mathbf{v} \in \mathbb{R}^{n \times 1}$, $\lambda_m(A)\|\mathbf{v}\|_2 \leq \|\mathbf{A}\mathbf{v}\|_2 \leq \lambda_M(A)\|\mathbf{v}\|_2$

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How are the quantized and full solution different?

$$\hat{\ell}^{\star} = \min_{\hat{\ell} \in \mathbb{R}^N} (\hat{\ell} - \mathbf{y})^{\top} \mathbf{C} (\hat{\ell} - \mathbf{y}) + \hat{\ell}^{\top} \mathbf{Q} \hat{\ell}$$

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We have: $\hat{\ell}_o = (\mathbf{C}^{-1} \hat{K}_o + \mathbf{I})^{-1} \mathbf{y}$ vs. $\hat{\ell}_{oq} = (\mathbf{C}^{-1} \tilde{K} + \mathbf{I})^{-1} \mathbf{y}$

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With linear algebra we get

$$\|\hat{\ell}_{oq} - \hat{\ell}_o\|_2 \leq \frac{\sqrt{n_I}}{c_u \gamma_g^2} \|\hat{K}_{oq} - \hat{K}_o\|_F$$

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Bounding quantization error $\left(\hat{f}_{soq,t}[t] - \hat{f}_{so,t}[t]\right)^2$

The quantization error depends on $\|\hat{K}_{oq} - \hat{K}_o\|_F = \|\hat{L}_{oq} - \hat{L}_o\|_F$.

When can we keep $\|\hat{L}_{oq} - \hat{L}_o\|_F$ under control?

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For what kind of data $\{\mathbf{x}_i\}_{i=1,\dots,n}$ is the distortion small?

Online SSL with Graphs: Analysis

Bounding quantization error $\left(\hat{f}_{soq,t}[t] - \hat{f}_{so,t}[t]\right)^2$

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Assume manifold $\mathcal{M}$

- all $\{\mathbf{x}_i\}_{i \geq 1}$ lie on a smooth d -dimensional compact $\mathcal{M}$
- with boundary of bounded geometry Def. 11 of Hein **hein2007graph**
 - has finite volume V
 - has finite surface area A
 - should not intersect itself
 - should not fold back onto itself

Online SSL with Graphs: Analysis

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Bounding $\|\hat{L}_{oq} - \hat{L}_o\|_F$ when $\mathbf{x}_i \in \mathcal{M}$

Consider k -sphere packing^{*} of radius r with centers contained in $\mathcal{M}$. *only the centers are packed, not necessarily the entire ball

If k is large $\rightarrow r < \text{injectivity radius of } \mathcal{M}$ hein2007graph

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r -packing is a $2r$ -covering:

$$\max_{i=1,\dots,N} \|\mathbf{x}_i - \mathbf{c}\|_2 \leq R \frac{R}{R-1} \leq 2\mathcal{O}\left(k^{-1/d}\right) = \mathcal{O}\left(k^{-1/d}\right)$$

But what about $\|\hat{L}_{oq} - \hat{L}_o\|_F$?

Online SSL with Graphs: Analysis

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Online SSL with Graphs: Analysis

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Online SSL with Graphs: Analysis

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Are the assumptions reasonable?

Online SSL with Graphs: Analysis

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$$\frac{1}{N} \sum_{t=1}^N (\hat{f}_{soq,t}[t] - \hat{f}_{so,t}[t])^2 \leq \frac{n_l}{c_u^2 \gamma_g^4} \|\hat{L}_{oq} - \hat{L}_o\|_F^2$$

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What does that mean?

Michal Valko

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Isara Labs



<https://misovalko.github.io/mva-ml-graphs.html>